

Control Engineering Lab Report

A report submitted to The University of Manchester for the degree of
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1 Introduction

This report is for the Control Engineering unit and the lab will use Matlab and Simulink to demonstrate the modelling and controller design for the Quanser Aero 2. It will provide an example of a cascade control structure for a single degree of freedom PID controlled system. The yaw axis will be locked so that only the pitch angle can be controlled by the motors and this is done to ensure the use of a one degree of freedom system.

In the desired configuration for this lab, the Quanser Aero 2 has an actuator subsystem (controls the speed of the rotor propellers) and a pitch angle subsystem (controls the pitch angle position). The inner loop (rotor speed) of the block diagram is controlled by the setpoint, which in turn is controlled by the outer loop (pitch control).

The aim of this lab is to build real time code, run it and collect the results in order to gain a practical understanding of how PID controllers work.

2 Modelling The System

2.1 Task 1

2.1.1 Take a screenshot of your complete Simulink model and discuss and justify it in your report

From the equations given in the lab instructions for this task, an equation for ω_m can be derived in terms of V_m , as follows:

$$J_{eq}\dot{\omega}_m(t) = \tau_m(t) - \tau_d(t) \quad (1)$$

where:

$$\tau_m(t) = k_t i_m(t) \quad (2)$$

$$\tau_d(t) = k_d \omega_m(t) \quad (3)$$

$$i_m(t) = \frac{v_m(t) - k_m \omega_m(t)}{R_m} \quad (4)$$

Inputting $\tau_m(t)$, $\tau_d(t)$ and $i_m(t)$ into the original equation and dividing both sides by J_{eq} gives:

$$\dot{\omega}_m(t) = \frac{\frac{k_t(v_m(t) - k_m \omega_m(t))}{R_m} - k_d \omega_m(t)}{J_{eq}} \quad (5)$$

Integrating the right hand side of this equation will result in an equation for $\omega_m(t)$ in terms of itself. This suggests that it is representative of a closed loop system, as once the output is acquired, it is input back into the equation as feedback in order to tune the final output value. This can be seen in the figure below:

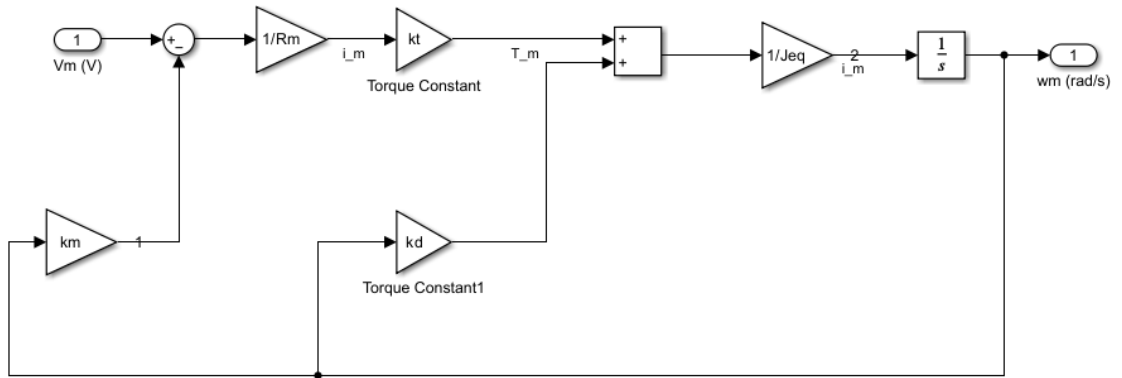


Figure 1: Simulink system block diagram for Quanser Aero 2

The Simulink model in Figure 1 shows an input of V_m for velocity and an output of ω_m for angular velocity. Through the feedback loop, $\omega_m(t)$ is multiplied by the drag/air resistance coefficient (k_d) gain block and the motor back emf constant (k_m) gain block separately, which can be expressed as $k_d \omega_m$ and $k_m \omega_m$ respectively. Using the circular sum block for V_m and the triangular gain blocks for torque constant (k_t) and the inverse of nominal resistance ($\frac{1}{R_m}$), the expression $\frac{v_m(t) - k_m \omega_m(t)}{R_m}$ is formed, which matches equation (4). Then, $k_d \omega_m$ is taken away from this, due to the use of the square sum block and multiplied by the triangular gain block for the inverse of the total moment of inertia ($\frac{1}{J_{eq}}$). At this point, the block diagram has outputted the expression on the right hand side of equation (5), with the final part being integrating this expression using the square integral block, labelled $\frac{1}{s}$.

The final output will be angular velocity and by translating the Simulink block diagram in Figure 1 to mathematical formulae, we are left with an equation to integrate with respect to t :

$$\dot{\omega}_m(t) = \frac{\frac{k_t(v_m(t) - k_m\omega_m(t))}{R_m} - k_d\omega_m(t)}{J_{eq}} \quad (6)$$

This matches equation (5) which was derived theoretically, proving the validity of the Simulink model.

2.1.2 Take a screenshot of all your scopes at all voltages and discuss in your report. Please discuss the observed results. Do these modelled systems represent the hardware correctly? What is the effect of changing the nominal setpoint? Explain.

The following graphs were obtained in the lab by running the system block diagram:

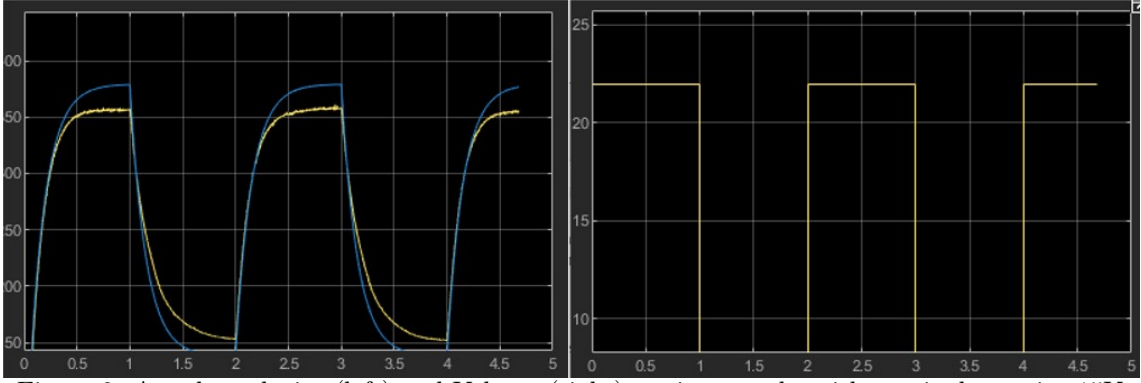


Figure 2: Angular velocity (left) and Voltage (right) vs time graphs with nominal setpoint 15V

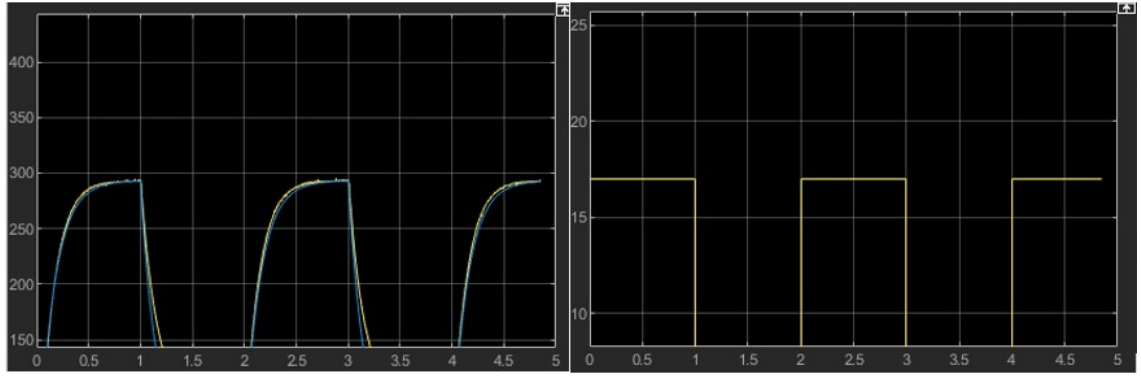


Figure 3: Angular velocity (left) and Voltage (right) vs time graphs with nominal setpoint 10V

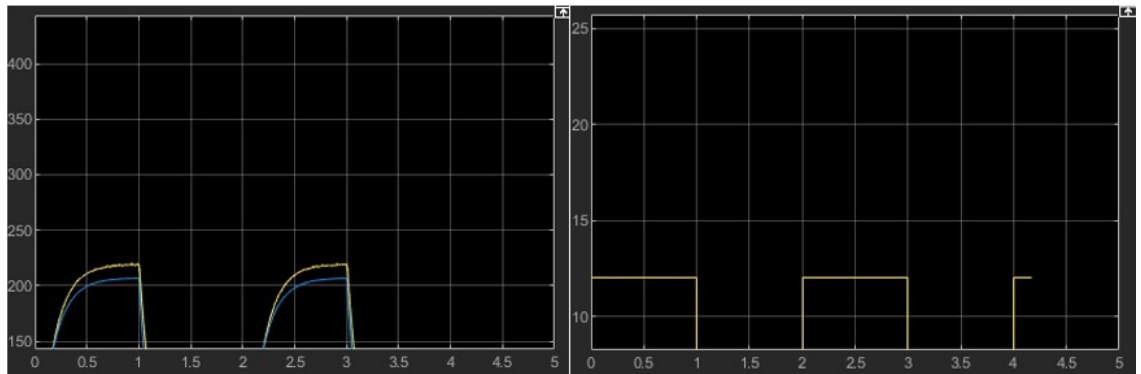


Figure 4: Angular velocity (left) and Voltage (right) vs time graphs with nominal setpoint 5V

In Figures 2, 3 and 4, the blue lines for angular velocity correspond to the modelled values, whereas the yellow lines correspond to the actual values produced and measured by the hardware. Because of this, it

can be seen that only at a nominal setpoint of 10V, the modelled system represents the hardware correctly because at 15V, the hardware undershoots and at 5V, the hardware overshoots. This could be due to the fact that a higher applied voltage results in a higher back emf constant. Back emf opposes applied voltage, meaning not as much angular speed was produced as expected, due to not considering a different value of k_m . It is the opposite situation for the graph at 5V, where the emf is not significant enough to override the voltage applied, leading to higher angular velocities than originally modelled. In general, the higher the nominal setpoint, the higher the angular velocity. These graphs are curved because the motor cannot instantaneously start or stop spinning, therefore there is a gradual increase/decrease in angular velocities with a decrease in rate of change. This aligns with what was physically observed in the lab.

The graphs on the right all show a clear uniform fluctuation from 0V to 30V, 20V, and 10V respectively. This is because when the hardware is on, it supplies a maximum voltage and when it is off, it supplies no voltage. It is always either on or off, which is why a square wave is formed. The ratio of voltage supplied when it is on to the voltage supplied when it is off equates to the average voltage (in this case, the nominal setpoint) which is directly related to the velocity of the motors. Decreasing the nominal setpoint will decrease the maximum voltage (i.e. the amplitude of the wave) because the average voltage (15V, 10V or 5V) has also decreased.

2.1.3 The Aero 2 thruster's motor shaft is connected to both the propeller and the propeller hub. Calculate the equivalent moment of inertia that is acting on the motor shaft. What assumptions can you infer about the effect of the propeller hub inertia on the system based on your working?

The equivalent moment of inertia that is acting on the motor shaft can be found by summing the rotor inertia, propeller hub inertia and propeller inertia, as follows:

$$J_{eq} = J_m + J_h + J_{prop} \quad (7)$$

Using the parameter values given in the lab assignment, this becomes:

$$J_{eq} = (4 \times 10^{-6}) + (3.04 \times 10^{-8}) + (4 \times 10^{-5}) = 4.40304 \times 10^{-5} \text{kgm}^2$$

Compared to the other two parameters, the propeller hub inertia has a much smaller value, making its effect on the value of equivalent moment of inertia much less significant. As described in the assignment, "the hub is a collet clamp used to mount the propeller to the motor" and it is evident that it will produce a much smaller inertia than the other components, as its dimensions are smaller, therefore we can assume that its effect is negligible.

2.2 Task 2

2.2.1 Plot the rotor speed response and motor voltage and discuss what these graphs are showing.

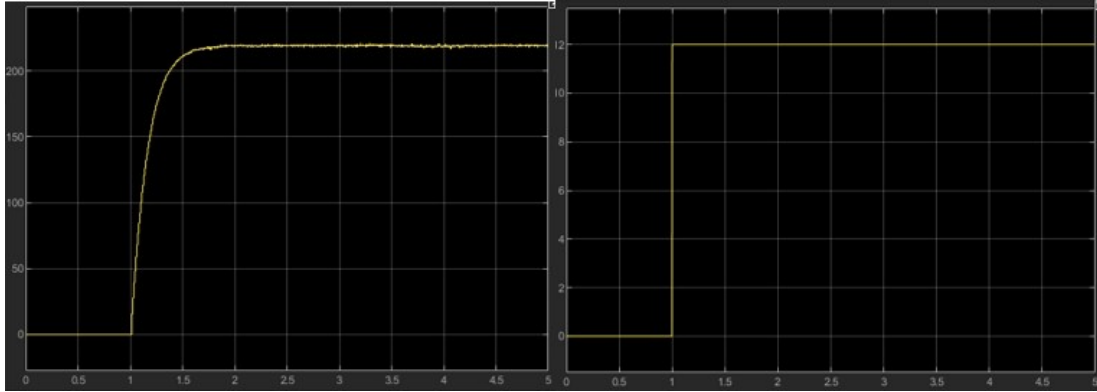


Figure 5: Angular velocity (left) and Motor voltage (right) vs time graphs

The graph on the right of Figure 5 shows the motor voltage produced and recorded by the hardware against time. As described in Task 1, the motors are always in one of two states: on or off. Because of this, the output shows that no voltage is applied for the first second of the plot and from 1 to 5 seconds (when the motor is spinning), 12V is applied. At the 1 second mark, the motor is turned on, causing a step in voltage which means it instantaneously increases from 0 to 12V.

The graph on the left of Figure 5 shows the angular velocity produced and recorded by the hardware against time, where the angular speed starts to increase at the 1 second mark, when the motor has been turned on. This is different to the other graph, as angular velocity cannot immediately jump to the required value, as the motor physically must gain speed. Because of this, a convergence to the final angular velocity value of around 220 rad/s can be seen at around the 2 second mark. Once this maximum value is reached, the graph shows miniscule fluctuations about this value for the remaining 3 seconds.

2.2.2 Calculate the steady-state gain, K and the time constant, τ , of the system from these results. Using this, what is the resulting transfer function model of the rotor?

As provided in the lab assignment, the equation for steady-state gain is given by:

$$K = \frac{\Delta y}{\Delta u} \quad (8)$$

Here, Δy is the difference between the steady-state output value (y_{ss}) and the initial output value (y_0).

Δu is the difference between the maximum input value (u_{max}) and the minimum input value (u_{min}).

Using the results obtained from the graphs in Figure 5, these two values can be obtained as follows:

$$\Delta y = y_{ss} - y_0 = 220 - 0 = 220 \text{ rad/s}$$

$$\Delta u = u_{max} - u_{min} = 12 - 0 = 12 \text{ V}$$

Substituting these into equation (8) gives the value of steady-state gain as:

$$K = \frac{220}{12} = 18.3$$

The equation for the time constant is given by:

$$\tau = t_1 - t_0 \quad (9)$$

Here, t_0 is the time when the input is applied to the system, which is 1 in this case. In the lab, the value of t_1 was found in a practical way by using the cursor measurements on the graph produced by Simulink. The point at which the system had reached roughly 63.2% of its output value was identified and the corresponding x axis value was read off the graph, with a value of 1.162 being recorded:

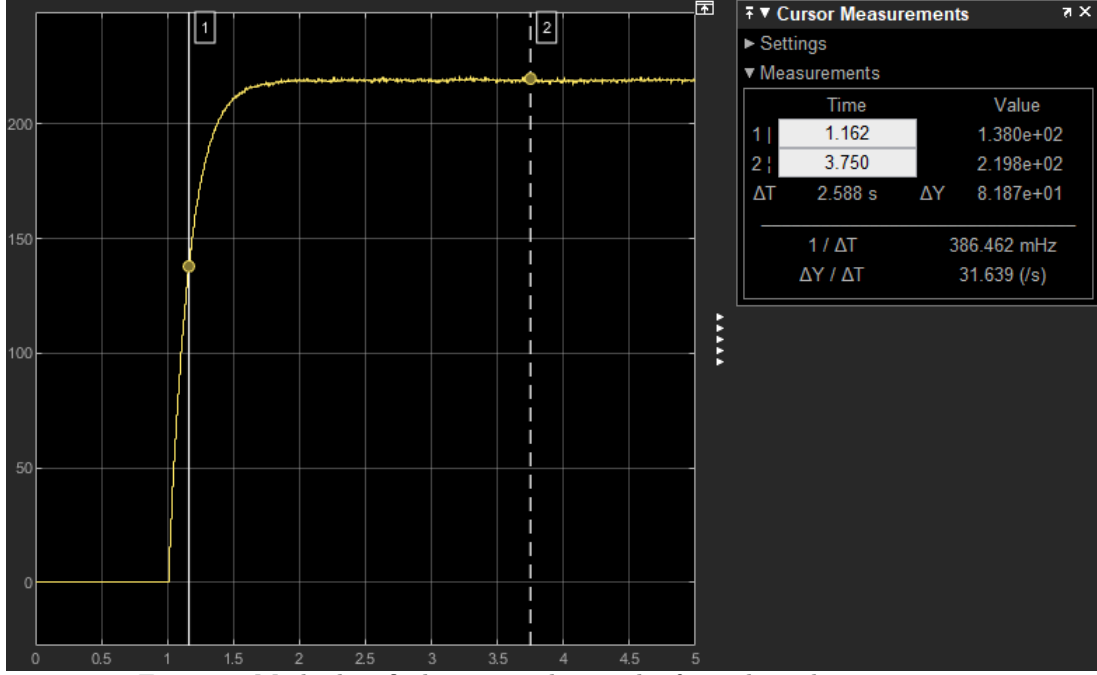


Figure 6: Method to find t_1 using the graph of angular velocity vs time

There is also a numerical way to obtain a value for t_1 , which has a greater chance of being more accurate than the above method. The equation for the output as a function of a given input is given by:

$$y(t_1) = (1 - e^{-1})\Delta y + y_0 = 0.632\Delta y + y_0 \quad (10)$$

Substituting the previously obtained values for Δy and y_0 gives:

$$y(t_1) = 0.632(220) + 0 = 139.04$$

Using interpolation, the value of t_1 that corresponds to a value of Δy of 139.04 was found to be 1.165, which is similar to the less accurate graphical value obtained above.

Substituting this value into equation (9) gives the following value for the time constant:

$$\tau = 1.165 - 1 = 0.165$$

Lastly, the transfer function model of the rotor speed can be expressed as:

$$\frac{Y(s)}{U(s)} = \frac{K}{\tau s + 1} \quad (11)$$

Substituting the calculated values of K and τ into the right hand side of equation (11) outputs the following transfer function for the rotor:

$$\frac{18.3}{0.165s + 1}$$

2.2.3 Attach a plot showing the response of the rotor speed of both the hardware and the model and the motor rotor voltage. What do these results indicate? Does this model represent the system well? Explain.

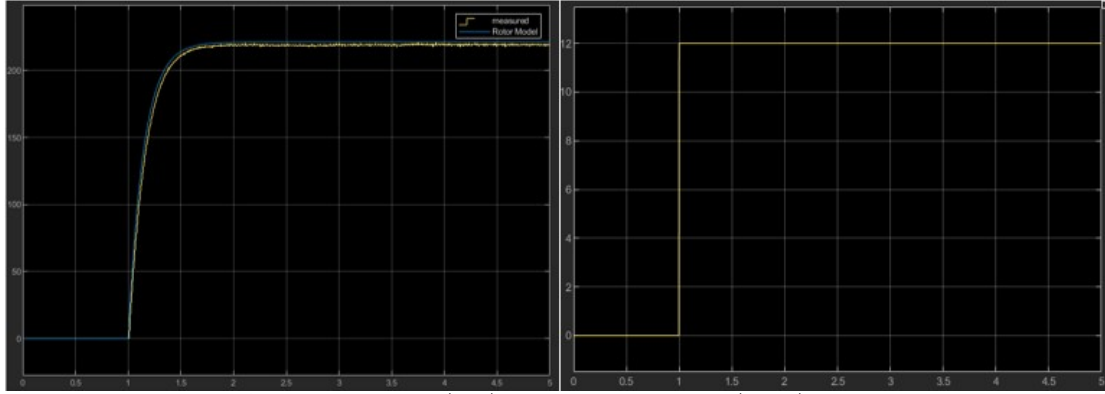


Figure 7: Angular velocity (left) and Motor voltage (right) vs time graphs

For the graph on the right (motor voltage vs time), the results are the same as for the first part of this task, picturing a step function that increases instantaneously from 0V to 12V at the 1 second mark.

For the graph on the left (angular velocity vs time), the blue line depicts the response of the rotor speed of the model and the yellow line is for the response of the rotor speed of the hardware. These lines were plotted on the same graph to allow easy comparison between the model and the actual system.

The results indicate that the model represents the system very well because the yellow line almost perfectly coincides with the blue line. However, the model slightly overshoot the converged value of angular velocity because the graph shows that the yellow line converges at a value that is very slightly less than that of the blue line, since the yellow line is a bit beneath it. This could be due to the emf opposing the applied input voltage, causing a negligible decrease in average angular velocity, however, the effect of this is insignificant at this value of voltage and the overall modelled results are represented well.

2.2.4 If it took twice as long for the Aero 2 to reach 63.2% of the steady-state speed, how would you have to change the transfer function model?

This change would affect the value of t_1 and therefore τ :

$$t_1 \times 2 = 1.165 \times 2 = 2.33$$

$$\tau = t_1 - t_0 = 2.33 - 1 = 1.33$$

The value of K would remain unchanged, leaving the new transfer function to be given as follows:

$$\frac{Y(s)}{U(s)} = \frac{18.3}{1.33s + 1} \quad (12)$$

2.2.5 If after applying the same input to the Aero 2, the steady-state speed was twice what you observed, how would you have to change the transfer function model?

Since the input is the same as Aero 2, t_1 and therefore τ will also remain the same. Changing the steady-state speed will only affect the values of y_{ss} , Δy and K :

$$y_{ss} \times 2 = 220 \times 2 = 440$$

$$\Delta y = y_{ss} - y_0 = 440 - 0 = 440$$

$$K = \frac{440}{12} = 36.6$$

This results in the following transfer function model:

$$\frac{Y(s)}{U(s)} = \frac{36.6}{0.165s + 1} \quad (13)$$

2.2.6 Based on your results, is the rotor system stable? Explain.

The results shown in the graph on the left in Figure 7 showed minor fluctuations for the yellow line even once the angular velocity value converged to about 220 rad/s and it is not a perfectly smooth line like the modelled blue one. This is representative of a real life rotor system, as the signal may be affected by system noise and several other factors. The fluctuations observed are so small, they can be assumed to be negligible. Because of this and the fact that convergence is achieved, the rotor system can be described as stable.

3 Controlling The System

3.1 Task 3

3.1.1 Based on your results, what can you observe on the system response when changing the proportional gain? Using your observations in step 5, do you think it is practical to use a proportional controller? Explain.

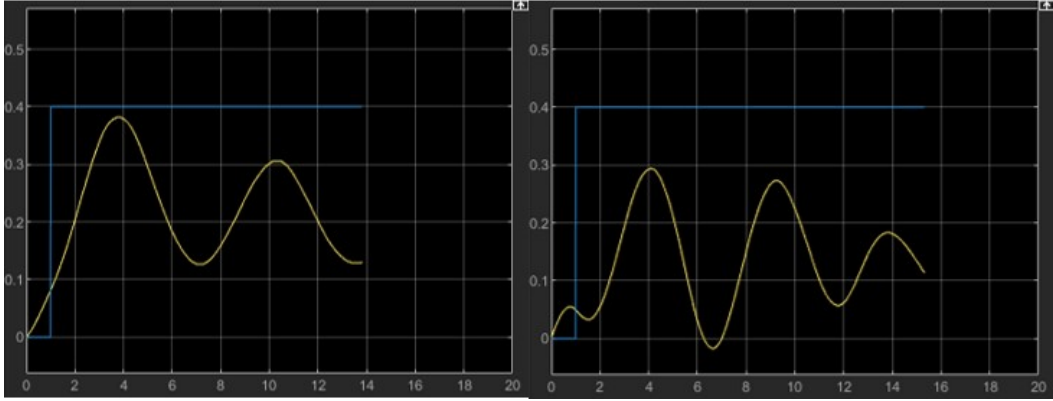


Figure 8: Pitch angle (rad) vs time (s) graphs with $k_p = 1, k_i = 0, k_d = 0$ (left) and $k_p = 4, k_i = 0, k_d = 0$ (right)

When the proportional gain was low, the system response was an undershoot, showing that it never even reached the modelled value (blue line in Figure 8), which was the desired pitch angle in radians. As it increased from 1 to 4, the peak time stayed roughly the same, however, more oscillations were observed for the same time and the peak pitch decreased. In both graphs, it can be seen that the system response begins to converge slowly, however, the steady-state error is very large.

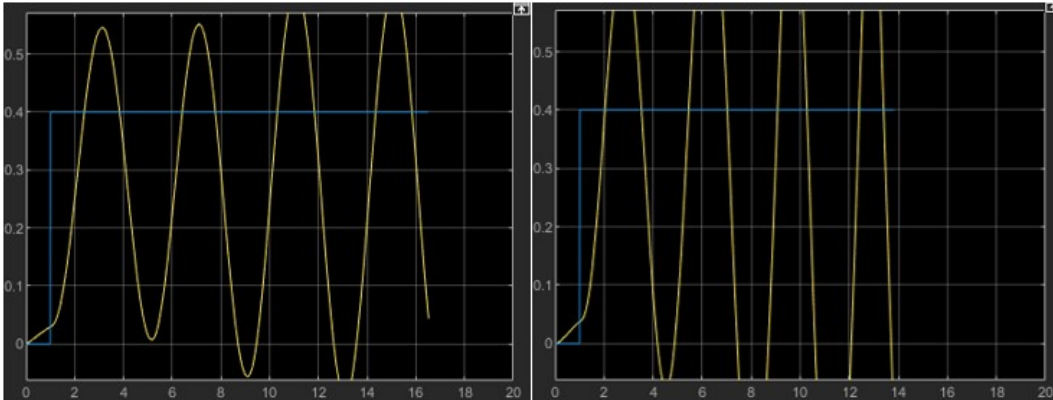


Figure 9: Pitch angle (rad) vs time (s) graphs with $k_p = 9, k_i = 0, k_d = 0$ (left) and $k_p = 15, k_i = 0, k_d = 0$ (right)

The proportional gain was increased and more oscillations occurred within the same time period, however, the severity of the overshoot also increased. It was difficult to observe any convergence for these values for such a short time frame, making it impossible to calculate any steady-state error, unless the system was monitored for a much longer period. This is because the greater the overshoot, the greater the current error, meaning the system will require more time to control the response and converge at the desired setpoint. The key characteristics of the graphs in Figure 9 is that they initially reach the desired pitch angle value more quickly as k_p is increased.

Overall, it is clear that it is practical to use a proportional controller because it tunes the system based on the difference between the current and desired positions. This means that without it, there is no guarantee that the system will ever reach the desired setpoint at all.

3.1.2 Based on your results, what can you observe on the system response when changing the derivative gain? What is the effect on the system when the derivative gain is very small. Answer based on your observed results in step 10.

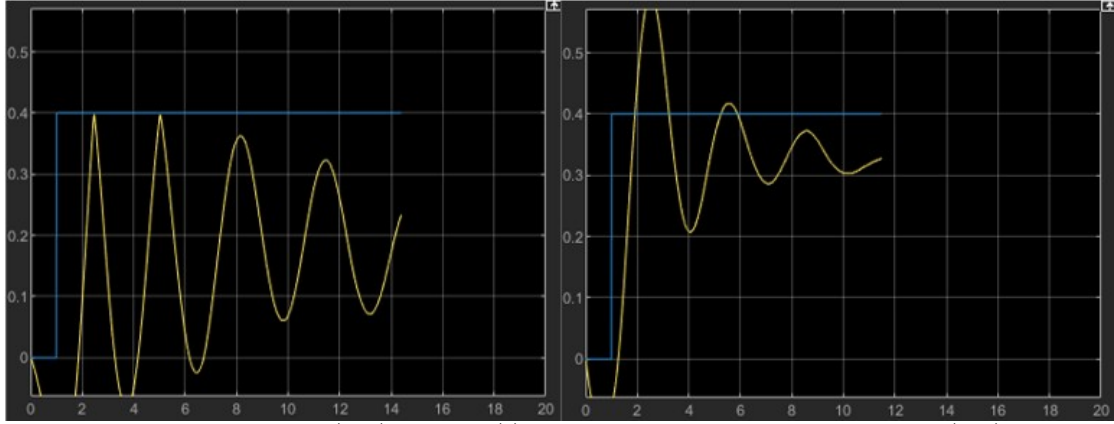


Figure 10: Pitch angle (rad) vs time (s) graphs with $k_p = 15, k_i = 0, k_d = 2$ (left) and $k_p = 15, k_i = 0, k_d = 5$ (right)

When the derivative gain was very small, the system response barely reached the desired setpoint and there was no overshoot at all, as seen in the graph on the left in Figure 10. The derivative gain was increased slightly and then overshoot and a decrease in steady-state error were observed, however converging was still slower than ideal. Despite still having a high value of k_p , introducing k_d caused a decrease in the rate of change of the system response.

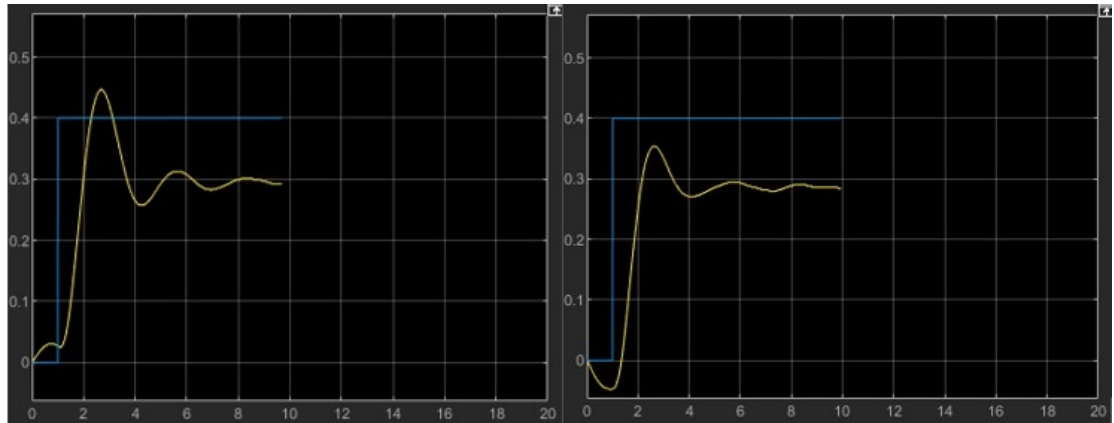


Figure 11: Pitch angle (rad) vs time (s) graphs with $k_p = 15, k_i = 0, k_d = 7$ (left) and $k_p = 15, k_i = 0, k_d = 10$ (right)

The minimum value of derivative gain that produced an acceptable overshoot (<0.45 radians) was found to be 7, as pictured on the left in Figure 11. As can be seen, higher values of k_d cause a significant damping effect on the system response, however if the values were too high, undershoot was observed and the system converged before the setpoint was ever reached. Changing the derivative gain had no effect on the peak time because it only affects the overshoot and size of the oscillations. One thing that was consistently observed was that at no point did the system response settle close enough as required to the setpoint.

3.1.3 Based on your results, what can you observe on the system response when changing the integral gain? What is the effect on the system when the integral gain is very large. Answer based on your observed results in step 11.

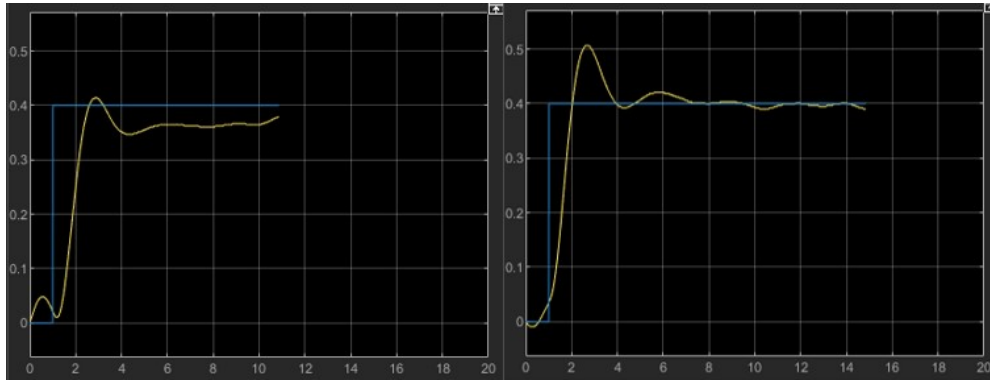


Figure 12: Pitch angle (rad) vs time (s) graphs with $k_p = 15, k_i = 3, k_d = 10$ (left) and $k_p = 15, k_i = 7, k_d = 10$ (right)

Based on previous errors, the integral gain was able to decrease the offset of the convergence that the system was producing when changing the proportional and derivative gains. As pictured on both graphs in Figure 12, adjusting the k_i value allowed the response to meet all of the system requirements and the values shown converged to within the desired range for the steady-state error (0.388-0.412 radians). When the integral gain was very large, the overshoot became too high and the settling time increased, as fluctuations about the setpoint were observed for longer. This is because k_i is proportional to past errors and fluctuations. However, when the integral gain was too small, the system response did not converge to a pitch angle value that was within the required range.

3.1.4 Plot the resulting system response overlaid on the command waveform and linear representations of the peak overshoot limits from Step 12. Record the final PID gains and comment on how you modified your controller to arrive at the results.

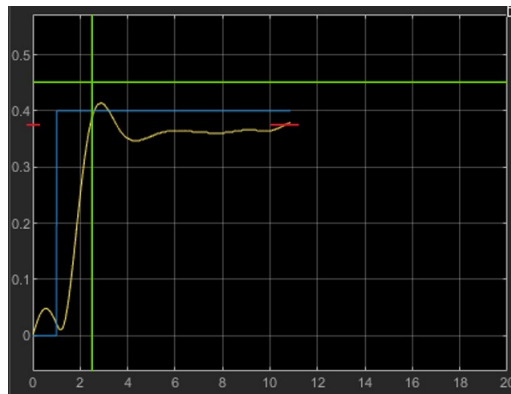


Figure 13: Pitch angle (rad) vs time (s) graph with $k_p = 15, k_i = 3, k_d = 10$

Initially, k_p was varied from 0-15 until the system reached the desired setpoint and was able to sustain oscillations. Then, k_d was varied from 0-10 until the rate of change in magnitude of the fluctuations decreased sufficiently to output a shorter settling time. Lastly, k_i was varied from 0-10 until the results met all of the requirements detailed in the assignment. In Figure 13, the red line highlights that the results converge to the desired range derived from the steady-state error, as it is within 3% of the desired setpoint (0.388-0.412 radians). The horizontal green line highlights that the peak pitch (0.45 radians) was not exceeded, as desired. The vertical green line highlights the peak time to reach the desired setpoint (2.2 seconds) was matched perfectly. Other combinations of these values were tested, yet none of them produced results as in Figure 13 that met all three limits set in this assignment.

4 Overall Conclusions And Reflections

4.1 Task 4

4.1.1 Please give a brief overview of PID control, including the following: Origin of the Algorithm, Description of the algorithm and its Ideal and Standard Forms, Benefits and Limitations, Potential improvements, References.

The first PID controller was developed in 1911 by Elmer Sperry (O'Dwyer 2005) and it was used to automate a ship's steering mechanism. However, the first theoretical paper on PID control was published in 1922 by Nicolas Minorsky. Minorsky was employed by the US Navy to develop a system to automatically steer ships and this was based on current error, past error and rate of change (Ho 2020). After World War 2, PID controllers became highly implemented, as at this point, ones with electrical systems had been developed.

PID control is an iterative feedback loop that allows a process to be controlled and maintained. The algorithm has a standard form and an ideal form that was developed to allow for a more practical implementation:

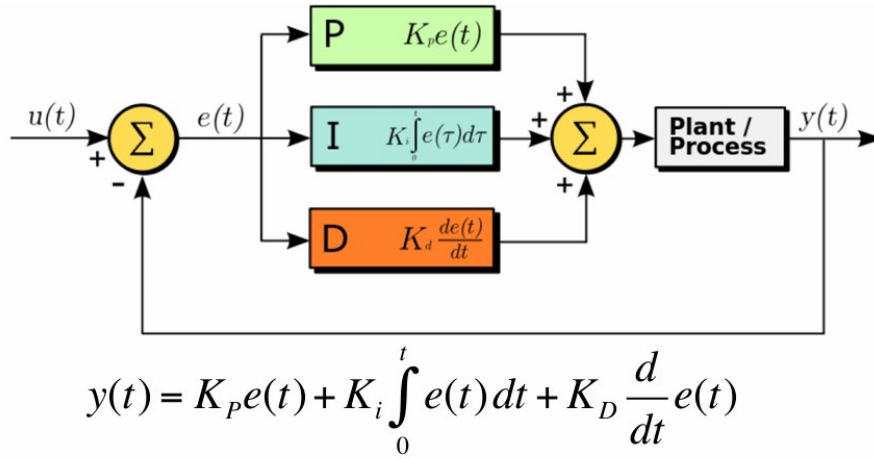


Figure 14: Ideal form block diagram of the PID algorithm (Abram n.d.)

In the ideal form, the three components that govern the control element of the process (proportional, integral and derivative) are independent and can therefore be adjusted separately.

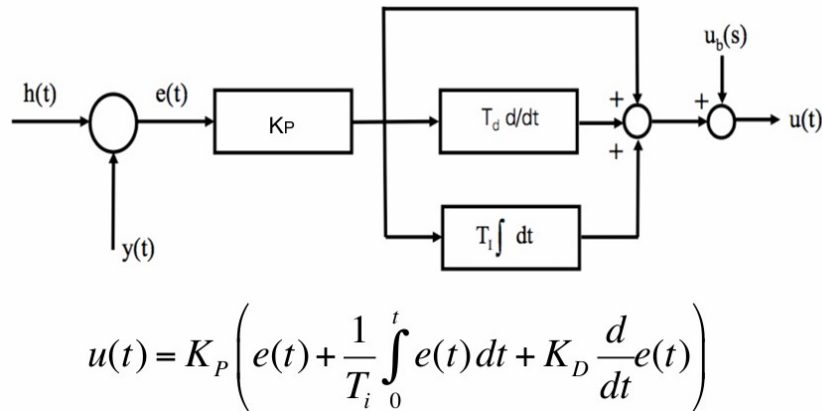


Figure 15: Standard form block diagram of the PID algorithm (Abram n.d.)

In the standard form, the integral and derivative components depend on the proportional component. This is because changing k_p tends to have the most significant effect on the system and therefore the ability to exclusively change one component more easily is desirable.

In general, PID is an algorithm that is used by a system to evaluate a certain variable that needs to be controlled. The initial value of this variable is measured and this information is sent to the controller via a feedback loop. The feedback signal is then compared to the desired setpoint and from this difference, an error is derived. Using the proportional, integral and derivative components of the algorithm, the controlled variable is then corrected in order to reduce the error.

The proportional term adjusts the error, the integral term removes the offset of the error and eliminates sluggish behaviour, the derivative term decreases the magnitude of the overshoot and the rate of change of the output.

The advantages of PID control include:

- It offers increased robustness and more precise control compared to most other types of controllers because they can constantly be adjusted to match the desired output.
- Its ability to remove the issue of over/undershooting allows these systems to significantly reduce energy consumption, making them more cost-effective.
- PID controllers can be used in many different industries, as they can be applied to a range of systems because of their flexibility and versatility.
- They are simple and easy to use and can therefore intuitively be tuned by engineers.
- They provide a measure of anticipation, meaning they can respond well and quickly to unexpected disturbances.
- They also provide excellent stability and have been used for a long time. The fact that they are so widespread means that they offer a level of consistency across systems that make them easier to troubleshoot and maintain.

The limitations of PID control include:

- PID controllers are highly sensitive to the errors produced by signal noise and measurements and this can amplify the components, leading to instability.
- It is a linear system, therefore has poor performance when the dynamics of a system change unpredictably.
- The integral component of the algorithm might delay the settling time due to a large build up of delays over a long period of time.
- It is not based on a physical model of the system, therefore it cannot be used to control implicit parameters.

Potential improvements to PID control (Astrom and Hagglund 2001) include:

- Replacing PID controllers with general linear controllers of more than one degree. These would allow several inputs and outputs instead of just one.
- Filters could be added to the control system to manage oscillations better.
- Combine PID controllers with gain scheduling and implement auto-tuning. This will allow great performance in both linear and non-linear systems.
- Separate regulation and set point response so that they are done independently, thus feedback that affects set point response is not continuously varied by load disturbances.
- Increase feedback frequency which would result in a more precise response adjustment.

4.1.2 Write a conclusion summarising the key outcomes of this lab and the potential applications of PID control.

A crucial concept to understand from Task 1 was how to link the closed loop system designed in Simulink to the equations given. An equation for ω_m was derived in terms of itself, proving that for each output, there would be some feedback. Running the system block diagram based on the derived equation produced results that showed the rate of change of angular velocity and these hardware results were compared with the modelled results. This task demonstrated that for a higher nominal setpoint for voltage, the hardware undershot and for a lower setpoint, it overshot the modelled angular velocity. Task 1 also highlighted the effects of inertia and that the propeller hub inertia had a negligible effect on the equivalent moment of inertia compared to the other parameters.

Task 2 was assessing the stability of the rotor system and the effects of different responses on the transfer function model. One key outcome of this task was calculating the effect of changing the inputs and outputs of the model on a first order transfer function, which then produced an accurate model based on the system response. Although the results converged, another key outcome was that unlike the modelled response, the true system output would never fully settle at one constant value - there would be continuous oscillations, however these were so miniscule that the rotor system was still overall considered to be stable. Lastly, Task 3 integrated P, I and D control to demonstrate the effect on the system response when changing each component of the block diagram independently. The outcomes observed were that changing the proportional gain changed the amplitude and frequency of the oscillations, changing the derivative gain adjusted the over/undershoot and decreased settling time and changing the integral gain removed the offset from the limits set.

Overall, these key outcomes showed how outputs are used on closed loop systems as feedback for tuning and that PID controllers can use errors to adjust the system.

Potential applications of PID control (Chen 2024) include:

- Regulating temperature, pressure and flow in the chemical industry.
- Regulating voltage and frequency in the power system industry.
- Reactive power compensation to enhance power system efficiency.
- Regulating speed, gas mixing and mechanical positioning in the manufacturing industry.

4.2 Task 5

- 4.2.1 Please use a small paragraph to summarize the teamwork in the lab sessions and the lab setup. Reflect on which part of the teamwork and the setup can be improved. What have you learnt from this lab and how can this benefit you in the future? What do you feel like you would improve about this lab and your approach to it for next time?**

I was fortunate enough to have this lab at the same time as my friends so finding a group and working efficiently was not an issue for us. We chose the group based on our strengths and weaknesses and formed our trio by recognising that we each had different strengths. This allowed us to work well as a team, as each person had a set role for each task and because of this, we were able to finish all of the tasks within our scheduled lab time. Although this was not an issue for us, I believe that deciding all groups beforehand would have been better, as there was a lot of chaos at the start of the lab which caused more organised groups to struggle to focus. It also meant that GTAs were busy helping people find groups instead of being available to answer questions.

This lab taught me the practical application of PID controllers, specifically the effects of changing each component of the algorithm on the system response. This will be extremely beneficial to me in the final exam of the Control Engineering unit because even though we are being taught the theory behind this subject, I believe that physically tuning the setup myself and plotting the results allowed me to directly compare the effects of changing the inputs and outputs.

The lab sheet was confusing at times because some tasks were removed, however the instructions on blackboard were not updated. This resulted in some unnecessary confusion during the lab, therefore I would improve the consistency of the lab sheets across all platforms.

If I were to do this lab again, I would have derived the formula for Task 1 beforehand, as this would have better prepared me to get a head start on the first task. I also would have familiarised myself with Simulink because this would have saved us time at the start of the lab.

References

- Abram, Tim (n.d.). “Lecture 2a - PID Controllers”. Slides 15-16.
- Astrom, K. J. and T. Hagglund (Apr. 2001). “The Future of PID Control”. In: *Elsevier*.
- Chen, Riu (2024). “A Comprehensive Analysis of PID Control Applications in Automation Systems: Current Trends and Future Directions”. In: *MCEE 2024*.
- Ho, Anthony K. (2020). *Fundamentals of PID control*. Tech. rep. PDH Online.
- O’Dwyer, Aidan (2005). “Control in the IT sector”. In: *PID Control: The Early Years*. Cork Institute of Technology.